

IFN655 |Advanced Business Intelligence Concepts for Enterprise Systems| ASSESSMENT #1 RUBRIC

Poor	Fair (4)	Good (5)	Very Good (6)	Excellent (7)	Marks
Group					
Understand the source data and generate a Data Profile for the given source dataset					
Demonstrates limited/no skills to develop a Comprehensive data profile and has the following issues. 1. The profile is incomplete 2. Fails to identify quality issues or misunderstands key aspects of data quality. Findings are poorly documented, and the visualizations are not meaningful.	Demonstrates a basic set of skills to develop a Comprehensive data profile by addressing the following. 1. Basic profiling conducted but missing key concepts (e.g., outliers, consistency) or Python code not 2. Identifies only some of the quality issues, leaving out significant ones. 3. Findings are documented with lack of depth of the explanations and visualizations and missing the clarity of them.	Demonstrates a good set of skills to develop a Comprehensive data profile by addressing the following. 1. considering an acceptable level key aspects such as (but not limited to) completeness, data types, formats, consistency, accuracy, outliers, data distribution, etc. 2. A good identification of quality issues that could affect the analysis but a few of them are missing 3. Findings are documented with some issues in depth of the explanations and visualizations where applicable	Demonstrates some advanced skills to develop a Comprehensive data profile by addressing the following 1. considering adequate number of key aspects such as (but not limited to) completeness, data types, formats, consistency, accuracy, outliers, data distribution, etc. 2. A very good identification of quality issues that could affect the analysis 3. Findings are mostly documented with minor issues in explanations and visualizations where applicable	Demonstrates Higher level. skills to develop a Comprehensive data profile by addressing the following 1. considering Most of the key aspects such as (but not limited to) completeness, data types, formats, consistency, accuracy, outliers, data distribution, etc. 2. Clearly identifies most of the quality issues that could affect the analysis 3. Findings are well documented with required explanations and visualizations where applicable	/15
Task 2 Perform Data Cleaning and Preparation					
Did not demonstrate the required set of skills to perform a data cleaning process. A poor decision-making process to address the quality issues from Task 1. Poor or no documentation on data cleaning process. Python code is not provided, or the provided code is mostly incomplete.	Demonstrates a basic set of skills to perform a data cleaning process by addressing some of the quality issues from Task 1. The data cleaning techniques were used but decision making to address the quality issues (such as treating missing values, erroneous values, duplicates) have concerns. A python code is provided with errors, and the report includes	Demonstrates an acceptable set of skills to perform a good data cleaning process by addressing an acceptable amount of the quality issues from Task 1. The data cleaning techniques were used and showed some acceptable decision making to address the quality issues (such as treating missing values, erroneous values, duplicates, A python code is provided with some issues, and the report	Demonstrates a good set of skills to perform a thorough data cleaning process by addressing most of the quality issues from Task 1. The data cleaning techniques were used appropriately and showed a good decision making to address the quality issues (such as treating missing values, erroneous values, duplicates, A python code is provided with minor issues, and the report	Demonstrates advanced skills to perform a thorough data cleaning process by addressing all quality issues from Task 1. The data cleaning techniques were effectively used showed and excellent decision making to address the quality issues (such as treating missing values, erroneous values, duplicates,	/12

	some explanation of the data cleaning process.	includes an acceptable explanation of the data cleaning process.	includes a good explanation of the data cleaning process.	A python code is provided, and it is error free, and the report includes an excellent explanation of the data cleaning process.	
Task 3 Load the Data to Azure Storage for ETL					
The data can not be loaded to the Azure storage or and incorrect formats were used for the data loading process. The report has no/limited set of evidence for data loading.	The data can not be fully loaded to the Azure storage and only a part of the data is successfully loaded to the storage. The file formats were not correctly considered, and the report has only a few evidence for the loading the data to the Azure storage.	The data is loaded to the Azure storage. But there are significant problems with correct use of file formats. The report shows some evidence for loading the data to Azure storage.	The data is successfully loaded to the Azure storage. But there are some problems with correct use of file formats. The report shows a good set of evidence for loading the data to Azure storage.	Successfully loads data to Azure Storage, considering the optimal file formats. The report shows an excellent set of evidence for loading the data to Azure storage.	/6
Task 4 Design the Destination table structure for ETL					
The destination schema is poorly designed/ Not designed, and it has significant issues or missing key tables/constraints. SQL scripts are incomplete, non-functional, or poorly written Key constraints and relationships are not implemented.	Demonstrates some level of skills to achieve the following 1.The database schema is designed but it has some major issues such as missing relations. 2. Missing a set of required tables at the destination. 3. Basic SQL scripts are provided but contain errors or are incomplete 4. Relationships and Constraints are identified but they have significant issues.	Demonstrates an acceptable set of skills to achieve the following 1.The database schema is designed but has some issues with the design. 2. All required tables at the destination have physically implemented but they have some functional issues. 3. SQL scripts are functioning with major errors. 4. Some of the key constraints and relationships are correctly implemented.	Demonstrates a good set of skills to achieve the following 1.The database schema is well designed, logical and aligned with the client requirement but having minor issues. 2. All required tables at the destination have physically implemented with minor inconsistencies in data types. 3. SQL scripts are functioning with minor errors. 4. Majority of the key constraints and relationships are correctly implemented.	Demonstrates advanced skills to achieve the following 1.The database schema is well designed, logical and fully aligned with the client requirement. 2. All required tables at the destination have physically implemented with correct data types. 3. SQL scripts are functioning without any errors. 4. All key constraints and relationships are correctly implemented.	/15

Task 5- Perform the transformations					
Poor or No attempt on transformations. 1. No or barely implemented transformations. 2. ETL pipeline is incomplete, non-functional, or inefficient. 3. The data does not load to the destination. 4. SQL scripts (if used) are missing or non-functional. 5. Poor or no documentation of transformations	Demonstrates some level of skillset to achieve the following 1. Transformations incomplete and all required transformations are not available. 2. Basic ETL pipeline implemented but with errors or inefficiencies 3. The data loading is mostly incomplete or most of the data is missing. 4. Basic SQL scripts (if used) but contain errors or are incomplete. 5. The report only has some basic explanation about the transformation.	Demonstrates a good level of skillset to achieve the following 1. Transformations are designed, but they have errors. 2. ETL pipeline is designed correctly but it has some functional issues. 3. The data loading to the destination is successful but there are some issues with data loading. 4. All SQL scripts used for the transformation are provided where appropriate. But they have some notable issues. 5. An explanation of transformations with screenshots is available in the report. But some of the details are missing.	Demonstrates a very good level of skillset to achieve the following 1. Transformations were well designed, but they have minor errors. 2. ETL pipeline is designed correctly but it has minor functional issues. 3. The data loading to the destination is successful but there are minor issues with data loading. 4. All SQL scripts used for the transformation are provided where appropriate. But they have minor issues. 5. An explanation of all transformations with screenshots is available in the report.	Demonstrates advanced skills to achieve the following 1. Transformations were well designed with no errors. 2. ETL pipeline is well-designed and fully functional 3. The data loading to the destination is successful without any missing data in the pipeline. 4. All SQL scripts used for the transformation are provided where appropriate. 5. A good explanation of all transformations with screenshots is available in the report.	/30
Task 6- Generate Insights using Power BI					
Do not demonstrate / poorly demonstrate the following 1. Can not Connect to the data warehouse prototype (destination database) 2. A poor/no analysis is provided 3. No/poor visualizations in the dashboards	Demonstrates some level of understanding of the following 1. Connects to the data warehouse prototype (destination database) but could not query all the tables. 2. The analysis is provided, but it is missing some key considerations 3. Provides some level of Power BI dashboard/s with visualizations	Demonstrates a good understanding of the following 1. Connects to the data warehouse prototype (destination database) and queries data but the querying has issues, or they are incomplete. 2. Provides an acceptable analysis with clear insights into charging station utilization, geospatial data, etc 3. Provides appropriate Power BI dashboard/s with relevant, insightful visualizations	Demonstrates a very good understanding of the following 1. Connects to the data warehouse prototype (destination database) and queries data but the querying has minor issues. 2. Provides a good analysis with clear insights into charging station utilization, geospatial data, etc 3. Provides a quality and appropriate Power BI dashboard/s with relevant, insightful visualizations	Demonstrates advanced skills to achieve the following 1. Successfully connects to the data warehouse prototype (destination database) and efficiently queries data. 2. Provides a comprehensive analysis with clear insights into charging station utilization, geospatial data, etc 3. Provides a high quality and appropriate Power BI dashboard/s with relevant, insightful visualizations.	/18

Task 7 Provide a set of recommendations for the given scenario					
Do not demonstrate / poorly demonstrate the following 1.Recommendations are not given/ Recommendations are vague, unsupported, or misaligned with strategic goals. 2.There are no justifications or mentions related to the insights	Demonstrates a some understanding to achieve the following 1. Recommendations are basic and may lack strong data backing or alignment with strategic goal. 2. The justifications are not aligned with the insights.	Demonstrates a good understanding to achieve the following 1.Provides a set of Recommendations that are data-driven, actionable, and align with strategic goals effectively. But there are some recommendations missing or they are not aligned with actions or goals. 2. The justification shows a Some connection to insights but lack of proper connections	Demonstrates a very good understanding to achieve the following 1.Provides a set of Recommendations that are data-driven, actionable, and align with strategic goals effectively. But has minor error with alignment and actions. 2. The justification shows a good connection to insights	Demonstrates advanced skills to achieve the following 1.Provides a set of Recommendations that are data-driven, actionable, and align with strategic goals effectively. 2. The justification is well connected with the insights	/9
Task 8 Provide the report and presentation					
Poor presentation and demonstration of the solution that fails to show any the significant value of the enterprise solution and how it works Did not use given time limit but it is not utilized well.	Rudimentary report and demonstration of the solution using basic functionality and the data that shows some value of the enterprise solution and how it works. Used the given time limit but it is not utilized well.	A good, succinct, engaging, convincing report and demonstration of the solution showing all the criteria mentioned in the above sections of the rubric which leaves a few questions unanswered regarding the value of the enterprise solution and how it works. Used the given time limit effectively.	Well developed, succinct, engaging, convincing report and demonstration of the solution showing all the criteria mentioned in the above sections of the rubric which leaves a few questions unanswered regarding the value of the enterprise solution and how it works. Used the given time limit very effectively.	Highly developed, succinct, engaging, highly convincing report and demonstration of the data analytics solution showing all the criteria mentioned in the above sections of the rubric which leaves no questions unanswered regarding the value of the enterprise solution. Used the given time limit for maximum impact.	/6
Task 9 Data Quality Improvement Proposal					
The proposal is missing, incomplete, or demonstrates little understanding of data quality concepts. No clear identification of current dataset limitations, and no meaningful or	Rudimentary proposal, which identifies a few limitations in the current dataset but lacks depth or clarity. Recommendations for improvement are minimal,	A good, succinct, engaging, convincing proposal which clearly identifies several relevant dataset limitations and provides a reasonable set of recommendations for improvement. Suggestions include	Well developed, succinct, engaging, convincing proposal thoroughly analyses dataset limitations and provides well-considered, relevant, and feasible recommendations. Proposes multiple new data fields,	Highly developed, succinct, engaging, highly convincing proposal which provides an insightful, comprehensive, and innovative improvement plan. Identifies critical and	/9

realistic recommendations are provided	generic, or not well-justified. Limited connection between proposed changes and their potential business value.	some relevant new data fields and collection methods, with basic justification of their value. Minor gaps in detail or feasibility	appropriate collection methods, and potential technology integration (e.g., IoT). Demonstrates clear understanding of how improvements would enhance business decision-making.	nuanced dataset limitations, and proposes highly relevant, feasible, and forward-thinking solutions. Proposes a complete set of new data fields and collection methods, and potential technology integration (e.g., IoT). Recommendations are well-justified, incorporate industry best practices, and clearly connect to strategic business benefits. Includes consideration of scalability, timeliness, and sustainability of data collection.	
				Overall	/120
				Final Marks	/40